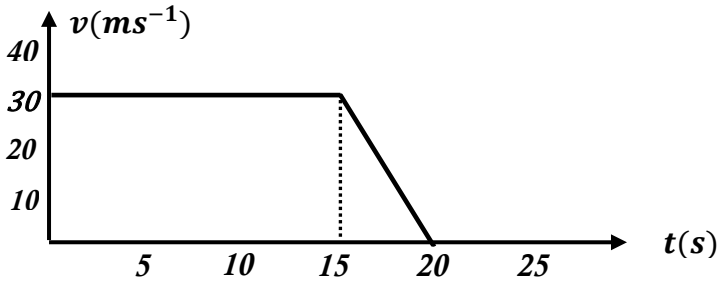
ESSLCE 2017/ (2025)

1. Which of the following correctly describes the difference between distance and displacement?
   A. Distance can be described both in magnitude and direction, while displacement is described only by magnitude.
   B. Distance depends on the initial and final points, while displacement depends on the path followed by a moving body.
   C. Distance is always greater than or equal to the magnitude of displacement, while the magnitude of the displacement is always less than or equal to distance.
   D. Distance has positive or negative values, while displacement has only positive values.
2. Which of the following statements is correct about a uniformly accelerated motion?
   A. The velocity is constant.
   B. The acceleration is constant.
   C. The speed is constant, but the direction is changing.
   D. The displacement increases at a
3. A train is moving along a straight line with a constant acceleration of 5 m/s22 when it passes by a traffic light point. From there on, if the time taken by the train to reach a velocity of 30 m/s is 4 seconds, then what is the distance of the train from the traffic point after 8 seconds?
   A. 80 m B. 160 m C. 240 m D. 320 m
4. An object moving in one dimension with constant acceleration a travels through a displacement s in a time interval t. If its velocity at the initial time t=0 is v_i and its velocity at a later time t is v_f, which of the following is a correct relation between the given quantities?
   A. $s = \left(\frac{v_f - v_i}{2}\right)t$ B. $s = v_i t + \frac{1}{2}at$ C. $s = \frac{v_f^2 - v_i^2}{a}$ D. $s = v_f t - \frac{1}{2}at^2$

ESSLCE 2016/ (2024)

1. A ball is thrown vertically upward with 12 m/s from the ground. Its speed when the ball is 4.0 m above the ground is:
   A. 15 m/s B. 12 m/s C. 8 m/s D. 10 m/s
2. A car travels 60 m north and then 80 m west. What is the magnitude of the resultant displacement of the car from the starting point?
   A. 20 m B. 4800 m C. 100 m D. 140 m
3. The velocity-time graph of an object moving along a straight line toward the east is described by the figure shown below. What is the total displacement moved by the object?
   A. 525 m East
   B. 600 m East
   C. 450 m East
   D. 375 m East

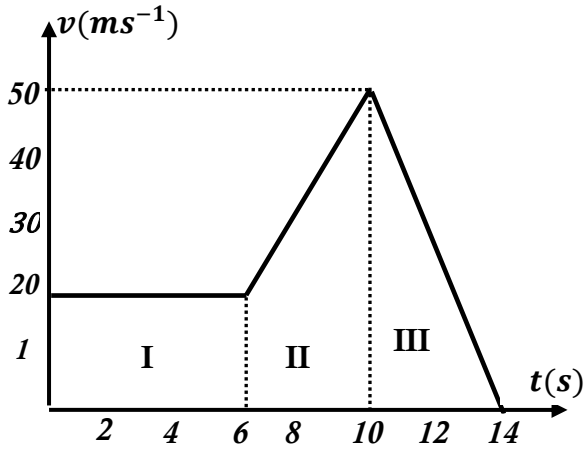


ESSLCE 2015/ (2023)

1. A boy on a journey covers his route by traveling 3.0 km east and 4.0 km north. What is the magnitude of his resultant displacement?
   A. 2.7 km B. 7.0 km C. 1.0 km D. 5.0 km

Tsega 2T

2. The graph given below shows the velocity of a man traveling on a motorcycle as a function of time. Which one of the following statements is correct about the velocity-time graph?
   A. The acceleration in the first 6 s is equal to the gradient of the curve in the given interval, and it is positive.
   B. The acceleration during the time interval from t = 6 s to t = 10 s is equal to the gradient of the curve in the given interval, and it is negative.
   C. The displacement during the time intervals t = 0 to t = 6 s equals the area under the velocity-time graph in the given interval.
   D. The displacement during the time interval from t = 6 s to t = 14 s equals the area of region Il minus the area of region Ill.



ESSLCE 2014/2015(2022)

1. For an object moving in a straight line,
   A. If the velocity increases by an equal amount in equal time intervals, the velocity-time graph is a straight line with a positive slope, and the acceleration-time graph is horizontal.
   B. The slope displacement-time graph and the region's area under the acceleration-time graph show the object's acceleration and velocity, respectively.
   C. The displacement-time graph is a horizontal line if the displacement changes by equal amounts in equal time intervals.
   D. The area of the region under the displacement-time graph and the slope of a velocity-time graph are, respectively, the acceleration and displacement of the object.
2. A student throws a ball vertically upward with a speed of 20 m/s. The maximum height reached by the ball and its velocity 3.0 s after it was thrown are, respectively:
   A. 20 m and 10 m/s downward.
   B. 20 m and 30 m/s upward.
   C. 15 m and 10 m/s downward.
   D. 105 m and 50 m/s upward.

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ESSLCE 2013/2014(2021)

1. A car is traveling at a constant velocity along a straight line. Which statement is correct?
   A. The displacement of the car is in the opposite direction.
   B. The displacement of the car increases linearly with time.
   C. The acceleration of the car is in the opposite direction.
   D. The net force on the car is different from zero.
2. Instantaneous velocity is different from the average velocity in that instantaneous velocity is:
   A. In the opposite direction to the displacement of the moving object.
   B. the velocity at a particular instant of time.
   C. the average velocity during the time of acceleration.
   D. the total displacement to the time taken.
3. A car traveling at 20 m/s starts to speed up. After 5 minutes, its speed is 144 m/s. What is the acceleration of the car?
   A. 0.41 m/s^2 B. 28.8 m/s^2 C. 0.55 m/s^2 D. 24.8 m/s^2

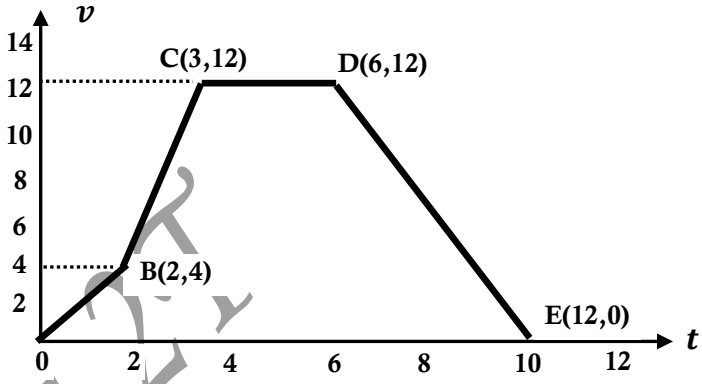
ESSLCE 2012/2013(2020/21)

1. While going to school, a student travels 3 km East and then 4 km North in 2 hrs. The magnitude of the student's average velocity and average speed are, respectively:
   A. 2.5 km/hr and 3.5 km/hr
   B. 2 km/hr and 7 km/hr
   C. 5 km/hr and 7 km/hr
   D. 3.5 km/hr and 2.5 km/hr

2. A boy throws a ball vertically upward with an initial speed of 20 m/s. Neglecting air resistance, which of the following is true about the motion of the ball?
   A. The time taken to reach the maximum height is 4s. **C.** Its maximum height is 20 m.
   B. Its velocity after 3s is 50 m/s, downward. **D.** Its velocity after 1s is 30 m/s, upward.

ESSLCE 2011(2019)

1. A driver of an automobile traveling at a constant speed of 20 m/s suddenly applies the brake, and the automobile comes to a stop in 2.0 seconds after skidding a certain distance. What is the length of the skid distance?
   A. 400 m **B.** 10 m **C.** 40 m **D.** 20 m
2. What is the magnitude of the resultant velocity for a bird flying first at a speed of 10 m/s northeast and then flying to South at a speed of 8 m/s?
   A. 7.13 m/s **B.** 16.6 m/s **C.** 12.8 m/s **D.** 7.07 m/s
3. Which of the following is correct about the motion shown in the velocity-time graph below?
   A. Between C and D, the motion is with constant acceleration.
   B. Between B and C, the acceleration is 4 m/s^2
   C. Between D and E, the motion is with constant positive acceleration.
   D. The total displacement is 84m.



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